

The inherent modality of verbs of perception, recognition, and disposition in Hungarian¹

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This paper is concerned with three verb classes in Hungarian: verbs of involuntary perception, verbs of involuntary recognition and verbs of physical disposition. Verbs falling into these natural classes pattern together across four, seemingly unrelated, phenomena: the incompatibility with an ability modal auxiliary, the existence of a dispositional middle voice form, the appearance in so-called root infinitives of circumstantial modality, and susceptibility to reinterpretation as auxiliaries. I argue that this is because of their unique, enriched semantics: such verbs have ability modality hard-wired into their lexical semantic representation. I will further argue that cross-linguistically, the lexical encoding of ability modality is an optional design element in the architecture of language which modulates the division of labour between (i) the various modules of language (syntax and semantics) and (ii) between speaker and hearer.

Keywords: ability modality; lexical modality; middle voice; root infinitives; grammaticalization

1. Introduction

This paper is concerned with three types of verbs:

- verbs of involuntary perception such as *lát* ‘see’, *hall* ‘hear’ or *érez* ‘feel’;
- verbs of involuntary (re)cognition such as *felismer* ‘recognize’ or *ért* ‘understand’;
- verbs of physical disposition such as *elér* ‘reach’ or *bír* ‘endure’.²

These 3 types of verbs pattern together across four, seemingly unrelated phenomena. Firstly, they cannot felicitously combine with the ability modal auxiliary *tud* ‘be_able_to’:

- (1) a. #*János tud-ja lát-ni az óceánt*
John be_able_to-3SG see-INF the ocean.ACC
a teraszáról.
the balcony.his.from
intended: ‘John can see the ocean from his balcony.’
- b. *János lát-ja az óceánt a teraszáról.*
John see-3SG the ocean.ACC the balcony.his.from
‘John can see the ocean from his balcony.’

Secondly, only a proper subset (see Appendix 1 and 2) of these verbs can appear in dispositional middles:³

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² A complete list of the verbs in question is provided in Appendix 1 and 2.

³ I.e., belonging to this verb class is a necessary but insufficient condition of appearing in a dispositional middle. The other conditions are discussed in detail in Section 4 below.

- (2) *Innen lát-sz-ik a hegycsúcs.*
 from.here see-MID-3SG the summit
 ‘The summit is visible to all from here.’

Thirdly, these (and only these) verbs can appear in so-called root infinitivals of circumstantial modality:

- (3) *Innen lát-ni a hegycsúcsot.*
 from.here see-INF the summit.ACC
 ‘One can see the summit from here.’

Finally, a subset of these verbs are such that they can function as ability modal auxiliaries themselves:

- (4) *Nem lát-ok olvas-ni.*
 not be_able_to_{visual}-1SG read-INF
 ‘I can’t read (because of a degradation of my ability to see).’

Based on these empirical observations, I will argue that in Hungarian, the verbs concerned have a unique and rich semantics: ability modality is hard-wired into their lexical semantics:⁴

- (5) $[[\text{lát}]]^{w,g} = \lambda x. \lambda y. \forall w' \in W [R_{\text{ability}}^y(w)(w') \rightarrow \text{see}^{w'}(y,x)]$

The semantic value of *lát* ‘see’ is function of type $\langle e, \langle e, t \rangle \rangle$, which returns the value true at possible world w iff it is the case that in every possible world w' that is consistent with the abilities of the denotation of the external argument (y), and where the latter is cognizant of their relevant abilities and chooses to exercise them, the latter sees the denotation of the internal argument. (Cf. Giannakidou 2001; Giannakidou–Mari 2021).

For clarity, this in-built lexical ability modality will also be reflected in the glosses (e.g. *lát* ‘BE_ABLE_TO.see’). I will argue that it is no coincidence that it is exactly this set of verbs that display in-built lexical modality. While in general, in-built lexical modality would lead to a problem of massive under- and overgeneration, with these particular verbs, this problem does not arise: since these verbs denote involuntary perception and (re)cognition, ability entails actuality (under the relevant scenarios). Finally, I will also explore how the lexical encoding of ability modality affects the division of labour between the various modules of language (syntax, semantics, pragmatics) and between speaker and hearer.

The paper is organized as follows. In Section 2, I discuss the incompatibility with the ability auxiliary *tud* ‘be able to’. In Section 3, I develop a new analysis of so-called ‘root infinitivals of circumstantial modality’. In Section 4, I introduce a novel proposal with regard to the semantics of dispositional middles. In Section 4, I discuss the subset of verbs of perception, (re)cognition and physical disposition which can be used as ability modal auxiliaries themselves. What unites these chapters is that the lexically in-built modality of the verbs under discussion form the centrepiece of the new analysis offered. In Section 6, I explore what the (non-)licensing of free-choice items reveals about the nature of lexically in-built ability modality. In Section 7, I offer an explanation to the question of why it is exactly this set of verbs that displays in-built lexical ability modality. Section 8 is the conclusion; which is followed by two appendices. Appendix 1 includes a tabulated overview of the syntactic behaviour of the verbs concerned, together with a short discussion. Appendix 2 includes a set of sentence samples illustrating each verb-structure combination.

⁴ This can be thought of as a special type of sublexical modality (Koenig & Davis 2001 and subsequent work).

2. Incompatibility with an ability modal auxiliary

Imagine the following scenario: a certain John owns a villa with a balcony that offers a view of the ocean. In English (and many other languages), this situation can be naturally described using an ability modal auxiliary:

(6) *John can see the ocean from his balcony.*

Unlike their counterparts in English, however, in Hungarian, verbs of involuntary perception, as well as verbs of involuntary (re)cognition and of physical disposition, cannot felicitously combine with an ability modal auxiliary (this has been observed by Kiefer (1984) wrt verbs of involuntary perception):

- (7) a. *#János tud-ja lát-ni aζ óceánt*
 John be_able_to-3SG BE_ABLE_TO.see-INF the ocean.ACC
a teraszáról.
 the balcony.his.from
 intended: 'John can see the ocean from his balcony.'
- b. *János lát-ja aζ óceánt a teraszáról.*
 John BE_ABLE_TO.see-3SG the ocean.ACC the balcony.his.from
 'John can see the ocean from his balcony.'
- (8) a. *#Nem tud-t-am fel-ismer-ni Jánost,*
 not be_able_to-PST-1SG PRT-BE_ABLE_TO.recognize-INF John.ACC
mert maszkban volt.
 because mask.in was
 intended: 'I could not recognize John because he was wearing a mask.'
- b. *Nem ismer-t-em fel Jánost,*
 not BE_ABLE_TO.know-PST-1SG PRT John.ACC
mert maszkban volt.
 because mask.in was
 'I could not recognize John because he was wearing a mask.'
- (9) a. *#Nem tud-om el-ér-ni*
 not be_able_to-1SG PRT-BE_ABLE_TO.reach-INF
a legfelső polcot.
 the highest shelf.ACC
 intended: 'I can't reach the highest shelf.'
- b. *Nem ér-em el a legfelső polcot.*
 not BE_ABLE_TO.reach-3SG PRT the highest shelf.ACC
 'I can't reach the highest shelf.'

The infelicitous versions are practically unattested in corpora (cf. Appendix 2) and are judged as very unnatural by native speakers. (Above, each verb class is represented by a single verb; cf. Appendix 1 and 2 for data on the full set of verbs.)

This pattern holds irrespective of whether the statement is generic or episodic. Consider first a scenario where Mary has high-diopter glasses. We can characterize her general situation the following way:

- (10) a. *#Mari nem tud lát-ni szemüveg nélkül.*
 Mary not be_able_to-3SG BE_ABLE_TO.see-INF glasses without
- b. *Mari nem lát szemüveg nélkül.*

Mary not BE_ABLE_TO.see-3SG glasses without

The second scenario is the following: I notice a friend at some distance, I call him on the phone and I keep waiving at him. Speaking through the phone, I ask him whether he can see me:

- (11) a. *#Tud-sz lát-ni engem?*
 be_able_to-2SG BE_ABLE_TO.see-INF me
 b. *Lát-sz engem?*
 BE_ABLE_TO.see-2SG me

Sentence (10) illustrates generic usage, whereas sentence (11) illustrates episodic usage; and in both cases, the verbs under discussion are incompatible with the ability modal auxiliary *tud* ‘be able to’. It is instructive to contrast this situation with English, where a different pattern emerges:

- (12) a. *Mary can’t see without her glasses.*
 b. *#Mary doesn’t see without her glasses.*
 intended meaning: ‘Mary lacks the ability to see without her glasses.’ (generic)
- (13) a. *Can you see me (right now)?*
 b. *Do you see me (right now)?*
 intended meaning: ‘Do you perceive me visually right now?’ (episodic)

In English, the ability modal auxiliary⁵ *can* may combine with a verb of involuntary perception. In a generic statement, an ability modal auxiliary is in fact obligatory if we want to obtain an ability modal reading. This suggests that in English, as opposed to Hungarian, the verbs under discussion do not have lexically specified ability modality.⁶

Turning back to Hungarian, I argue that the infelicitousness of these verbs with an ability modal auxiliary is due to redundancy: since ability modality is lexically specified, it is redundant to reintroduce it in the form of an ability modal auxiliary.

Thus, I propose that sentences with these verbs are always modal ability sentences in terms of their asserted meaning, and in the case of episodic (non-generic) readings, the non-modal interpretation (if any) is an actuality entailment (Bhatt 1999):

- (14) *Tegnap este lát-t-am a királynőt.*
 yesterday evening BE_ABLE_TO.see-PST-1SG the queen.ACC
 i. ASSERTION:
 ‘Yesterday evening I had the ability to visually perceive the queen.’
 ii. ENTAILMENT:
 ‘Yesterday evening I visually perceived the queen.’

With verbs of involuntary perception and (re)cognition, these entailments obtain independently of whether the verb is perfective (such as *felismer* ‘recognize’ in 8) or imperfective (such as *lát* ‘see’ in 7). This phenomenon is thus somewhat different from the well-known observation of actuality entailments associated with perfective ability modals (Bhatt 1999; Hacquard 2009).⁷

⁵ The auxiliary *can* of course has other modal readings too (such as epistemic or deontic), however, it is clear that in the context of (12) and (13), it has an ability reading. (This is also evidenced by the lack of interchangeability with *may* in 12 and 13).

⁶ *Recognize* and *understand* seem to pattern differently (cf. Appendix 2).

⁷ It is important to note that generic-episodic and imperfective-perfective are independent dimensions. Consider

(i) *János nem lát szemüveg nélkül.* GENERIC & IMPERFECTIVE
 John not BE_ABLE_TO.see glass w/o
 ‘John cannot see without his glasses.’

I argue that this pattern is due to fact that these verbs denote involuntary perception (note that (re)cognition is also perception on an abstract level): if you are able to see something, it follows that you actually see it (as long as the relevant internal and external conditions hold, see discussion below in Section 3).

As expected, verbs of *voluntary* perception (such as *néz* ‘look’) or (re)cognition (*fel-idez*⁸ ‘recall’) are indeed compatible with the modal *tud* ‘be_able_to’ (and without it, they cannot have an ability reading):

- (15) a. *Mari nagyon szigorúan tud-ott néz-ni.*
 Mary very sternly be_able_to-PST.3SG look-INF
 ‘Mary could look around very sternly.’
 = ‘Mary possessed in the past the ability to look around very sternly.’
- b. *Mari nagyon szigorúan néz-ett.*
 Mary very sternly look-PST.3SG
 = ‘Mary looked around very sternly (at a given point in time in the past).’
 → an ability modality reading is not accessible
- (16) a. *Mari fel tud-ta idéz-ni*
 Mary PRT be_able_to-PST.3SG recall-INF
a gyerekkori élményeit.
 the childhood experience.PL.ACC
 ‘Mary could recall her childhood experiences.’
 = ‘Mary possessed in the past the ability to recall her childhood experiences.’
- b. *Mari fel-idez-te a gyerekkori élményeit.*
 Mary PRT-recall-PST.3SG the childhood experience.PL.ACC
 = ‘Mary recalled (at given point in time in the past her childhood experiences).’
 → an ability modality reading is not accessible

Verbs of physical disposition fall into two classes. Verbs of the first class (such as *bír* ‘BE_ABLE_TO.endure’, *tűr* ‘BE_ABLE_TO.endure’, *győz* ‘BE_ABLE_TO.cope_with’, *áll* ‘BE_ABLE_TO.withstand’) denote a physical or mental ability (endurance, patience, stamina, resilience etc.). These verbs are thus involuntary, similarly to verbs of involuntary perception and (re)cognition, and possibility entails actuality here as well: if I am disposed to endure cold whether, this is physical ability of mine which I do not need to consciously, voluntarily exert or effectuate.

Verbs of the second class (such as *elér* ‘BE_ABLE_TO.reach’, *befér* ‘BE_ABLE_TO.fit_into’, *kifér* ‘BE_ABLE_TO.fit_throuh_on_way_out’, *átfér* ‘BE_ABLE_TO.fit_through’) denote the relative physical position or size of two entities. Here, possibility does not entail actuality: the fact that something is within my reach (i.e., I can reach it if I wish) does not entail that I actually reach for

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- (ii) *János mindenben meg-lát-ja* GENERIC & PERFECTIVE
 John all.INE PRT-BE_ABLE_TO.see-3SG
a jót.
 the good.ACC
 ‘John can recognize something good in everything.’
- (iii) *János tavaly decemberben, Londonban* EPISODIC & IMPERFECTIVE
 John last.year december.INE London.INE
látta az angol királyt.
 BE_ABLE_TO.see.PAST.3SG the English king.ACC
 ‘Last December, in London, John saw the King of England.’
- (iv) *János meg-lát-t-a a pisztolyt* EPISODIC & PERFECTIVE
 John PRT-BE_ABLE_TO.SEE-past-3SG the gun.ACC
Mari kezében.
 Mary hand.3SG.INE
 ‘John caught sight of the gun in Mary’s hands.’

⁸ Other involuntary-voluntary pairs show the exact same contrast (cf. Appendix 2).

it and grab it. In fact, I need to exercise voluntary action in order to actually reach and grab something that is within my reach. With such verbs, actuality entailments only obtain with perfectives (cf. Bhatt (1999), Hacquard (2009) a.o.):

- (17) a. IMPERFECTIVE READING
(Már 10 évesen) el-ér-t-em a legfelső polcot,
 already 10 year PRT-BE_ABLE_TO.reach-PST-1SG the top shelf.ACC
 ‘Already at the age of 10, I could reach the highest shelf,
 i. ASSERTION: ‘(already at the age of 10) I possessed the ability of reaching the highest shelf’
 ii. ACTUALITY ENTAILMENT: none
le tud-t-am volna venni bármit.
 PRT be_able_to-PST-1SG would pick-INF anything.ACC
 I could have picked off anything.’
- b. PERFECTIVE READING
El-ér-t-em a legfelső polcot
 PRT-BE_ABLE_TO.reach-PST-1SG the top shelf.ACC
 ‘I reached the highest shelf,
 i. ASSERTION: ‘I possessed the ability of reaching the highest shelf.’
 ii. ACTUALITY ENTAILMENT: ‘I actually did reach the highest shelf.’
és le-vet-t-em a lekvárt.
 and PRT-pick-PST-1SG the marmelade.ACC
 I picked off the jar of marmalade.’

(17a) has an imperfective reading, and as a consequence, no actuality entailment obtains. (17b) has a perfective reading, and as expected, the actuality entailment obtains. Note that the verbal particle *el* can be associated both with perfective and imperfective readings (Dékány 2008): in the examples above, we modulated the range of available readings by way of context (the second sentence).

Before concluding this section, we have to attend to two issues. So-called multiple modals are well-known from various languages: this is the phenomenon where one sentence contains two (or more) modal operators (Abraham 2001, Huang 2011):

- (18) *I don't think I have any grants you might could apply for.*
 (Di Paolo 1989)

In (18), the epistemic modal operator *might* has scope over the ability modal operator *can*. This phenomenon exists in Hungarian too:

- (19) *Esetleg Mari tud-bat segít-eni neked.*
 possibly Mary be_able_to-POSS-3SG help-INF 2SG.DAT
 ‘Mary may be able to help you, possibly.’ POSSIBILITY > ABILITY

Here, the possibility modal operator expounded by *-bat* has scope over the ability modal operator *tud*. In light of this, it is unsurprising that multiple modality can also be observed with regard to verbs with lexical ability modality:

- (20) *Esetleg Mari emlékez-het a PIN kód-ra.*
 possibly Mary BE_ABLE_TO.remember-POSS-3SG the PIN code-SUB
 ‘Mary may remember the PIN code.’ POSSIBILITY > ABILITY

In (20), the epistemic modal operator expounded by *-het* has scope over the lexically specified ability modal operator.

As we have discussed above, the incompatibility of the verbs under discussion with the modal auxiliary *tud* ‘be able to’ is due to redundancy avoidance. Since redundancy avoidance is a strong but by no means absolute tendency, our expectation is that it can be overridden under specific circumstances. This is indeed the case. Under the right circumstances, the verbs under discussion can combine with the copula + *képes* ‘able_{adj}’ quasi-modal construction, as well as with the ability modal auxiliary *tud* ‘be able to’:

- (21) a. *János a szembesítésen*
 John the lineup.SUP
nem ismer-t-e *fel az elkövetőt.*
 not BE_ABLE_TO.recognize-PST-3SG PRT the culprit.ACC
 ‘John could not recognize the culprit in the lineup.’
- b. *János a szembesítésen*
 John the lineup.SUP
nem volt képes fel-ismer-ni *az elkövetőt.*
 not was able BE_ABLE_TO.recognize-INF the culprit.ACC
 ‘John could not recognize the culprit in the lineup.’
- c. *János a szembesítésen*
 John the lineup.SUP
nem tud-t-a *fel-ismer-ni* *az elkövetőt.*
 not be_able_to-PST-3SG BE_ABLE_TO.recognize-INF the culprit.ACC
 ‘John could not recognize the culprit in the lineup.’

It is important to note that in general, no matter the verb, the copula + *képes* ‘able_{adj}’ construction is used when we discuss an ability being exercised to an extreme degree, describing an event where the ability in question was stretched to its limits:

- (22) a. *#A pisztráng képes Ø úsz-ni.*
 the trout able be.3SG swim-INF
 (be.3SG is phonologically null in Hungarian)
 Intended: ‘Trouts can swim (and this is a remarkable feat).’
- b. *A pisztráng tud úsz-ni.*
 the trout be_able_to-3SG swim-INF
 ‘Trouts can swim.’
- c. *A pisztráng képes Ø órákon át*
 the trout able be.3SG hour.PL.SUP through
az árral szemben úsz-ni.
 the current.INS against swim-INF
 ‘Trouts can swim against the current for hours (and this is a remarkable feat).’

Since swimming is an inherent ability of fish, the use of *képes* ‘able_{adj}’ is infelicitious in (22a), only (22b) sounds perfectly natural. (22c), on the other hand, is felicitous, since it describes a scenario where trouts exercise an inherent ability of theirs to an extreme degree.

(21) describes a very similar situation. In the lineup, John naturally exerted his ability of recognition, doing his very best to recognize the culprit. When describing such a situation, the speaker can follow one of three strategies. It is always possible to simply use *felismer* ‘BE_ABLE_TO.recognize’: this is fully adequate as the ability modality is lexically specified. However, if the speaker wishes to emphasise that John exercise his ability to a heightened degree, they may strategically employ redundancy either by using the copula + *képes* ‘able_{adj}’ construction (21b) or by using the ability modal auxiliary *tud* ‘be able to’ (21c). In each case, we end up with a redundant sentence containing two exponents of ability modality: a lexically specified one as well as one introduced in the course of syntactic derivation. This can be thought of an abstract instance of

contrastive reduplication (cf. *nagyon-nagyon* ‘very-very’, meaning ‘to an extreme degree’ or *cukor-cukor* ‘sugar-sugar’, meaning ‘real sugar (as opposed to some artificial sweetener)’, cf. Ghomeshi et al. 2004), which can in very general terms be described as the strategic deployment of redundancy in order to encode the extreme degree of some quality.

To conclude, for completeness, I briefly point out that the phenomenon of coercion exists here as well:

- (23) a. *Akármit mond-ok, a párom tudj-a*
 whatever.ACC say-1SG the pair.1SG be_able_to-3SG
úgy ért-eni, hogy az sértő legyen.
 so understand-INF that it insulting be.IMP.3SG
 ‘Whatever I say, my partner can interpret it as an insult.’
 b. *Akármit mond-ok, a párom képes Ø*
 whatever.ACC say-1SG the pair.1SG able be.3SG
úgy ért-eni, hogy az sértő legyen.
 so understand-INF that it insulting be.IMP.3SG
 ‘Whatever I say, my partner can interpret it as an insult.’

In (23), by coercion, the speaker uses *ért* ‘BE_ABLE_TO.understand’, a verb of involuntary perception, as a verb of voluntary perception (‘interpret in a specific way on purpose’). Under this coerced interpretation as a voluntary verb, *ért* can be freely combined with the auxiliary *tud* ‘be able to’ or the copula + *képes* ‘able_{adj}’ complex.

3. Root infinitives of circumstantial modality

So-called root infinitives of circumstantial modality have three main characteristics: (i) they function as independent root clauses, even though the main verb is non-finite (infinitival), (ii) they have a modal interpretation and (iii) they are interpreted as generic (Balogh 1890; Bartos 2002). Their modality is traditionally characterized as circumstantial (Bartos 2002), however, as we will see, the modality of these sentences can be straightforwardly analyzed as ability modality.

Strikingly, the verbs that participate in this construction are the exact same ones that cannot combine with the ability modal *tud* ‘be_able_to’ (further examples are provided in Appendix 1 and 2):

- (24) a. *Innen lát-ni a hegycsúcsot.*
 from.here BE_ABLE_TO.see-INF the summit.ACC
 ‘One can see the summit from here.’
 b. *A hangjáról rögtön fel-ismer-ni*
 the voice.3SG.DEL at.once PRT-BE_ABLE_TO.recognize-INF
János-t.
 John.ACC
 ‘One can readily recognize John based on his voice.’
 c. *Padovából könnyen el-ér-ni*
 Padova.ELA easily PRT-BE_ABLE_TO.reach-INF
mind a tengert, mind a hegyeket.
 both the sea.ACC both the mountains.ACC
 ‘From Padova, one can easily reach both the sea and the mountains.’

Root infinitivals of circumstantial modality denote propositions that are generic and modal at the same time: generally speaking, everyone has the ability to see the summit from here, to recognize John by his voice or to reach with ease both the sea and the mountains from Padova.

Bartos (2002) argues convincingly against a biclausal analysis (the idea that there is a silent matrix clause containing the modal operator which subsumes the root infinitive). However, since Bartos (2002) does not ascribe any inherent modality to the verbs concerned, he needs to stipulate a silent Mod operator right above CP:

$$(25) \quad \text{MOD}_{\text{circ}} [\text{CP} \dots [\text{VP} \dots \text{V}_{\text{inf}} \dots]]$$

This arrangement is somewhat unusual, as typically, CP is supposed to be the top layer of the clause. Bartos (2002) acknowledges the atypicality of his proposal by half-jokingly characterizing it as a sesqui-clausal (one-and-a-half-clausal) analysis midway between the bi-clausal and mono-clausal alternatives.

Bartos (2002) characterizes the set of verbs involved as follows: (i) the subject is inactive, (ii) there is some cognitive reaction/participation on behalf of the subject; (iii) the subject is a goal or a beneficiary. Dömötör (2025) reports a corpus analysis of this construction and characterizes the verbs concerned as non-agentive verbs of perception with an experiencer subject.

Both authors capture certain characteristics of the verbs concerned successfully (e.g. the role of involuntariness); however, the actual characterizations that they offer are either too broad or too narrow. The proposal of Bartos (2002) does not cover verbs of physical disposition, as with these verbs, there is no cognitive reaction or participation and the subject is neither a goal nor a beneficiary; rather, it appears to be a theme argument of a predicate describing a relative physical position or size. Talking of thematic roles, it also appears more appropriate to characterize the subject of verbs of perception as an experiencer (here, we are in agreement with Dömötör 2026).⁹ Dömötör (2026) provides a precise and empirically correct characterization of verbs of involuntary perception and recognition; however, her definition excludes verbs of physical disposition.

Both problems (i.e., the sesquiclausal syntax and the difficulties of defining the class of verbs that participate in the construction) disappear under my proposal: since the verbs involved are all inherently modal, there is no need for an extra-clausal silent modal operator and we can give a clear lexical semantic characterization of the verbs involved: if and only if a verb has lexically in-built ability modality, then it can participate in a root infinitival of circumstantial modality.

Before concluding this section, we have to attend to a matter of terminology. In this paper, we characterize the type of modality under discussion as ability modality; whereas Bartos (2002) ascribes circumstantial modality to root infinitivals. Is this a difference of substance or merely of terminology? It is instructive to first revisit here Kiefer's (2007: 322-324) discussion of the relationship between ability (which he takes to be a category related to but different from modality), circumstantial modality and dispositional modality:

- (26) a. *János nem tud zongoráz-ni.*
 John not be_able_to-3SG piano.play-INF
 'John cannot play the piano. (He never learnt to play it.)'
 b. *János most nem tud zongoráz-ni.*
 John now not be_able_to-3SG piano.play-INF
 'John cannot play the piano right now. (He is too upset to concentrate.)'
 c. *János itt nem tud zongoráz-ni.*
 John here not be_able_to-3SG piano.play-INF
 'John cannot play the piano here. (There is no piano at the premises.)'

Kiefer (2007) claims that in (26a), *tud* has a general ability reading (which Kiefer separates from modality strictu sensu). (26b) has a dispositional modal interpretation: while John in general has

⁹ There are only two verbs, *talál* 'BE_ABLE_TO.find' and *kap* 'BE_ABLE_TO.find', where the beneficiary analysis may appear viable; however, as we will see later on, these are also clearly verbs of perception (certainly in the constructions under discussion).

the ability to play the piano, in this very moment, she cannot play the piano as she lack the necessary disposition (physical or mental). (26c) has a circumstantial modal interpretation: while Anna in general possess the ability of piano playing, right now, external conditions make her exercise of these abilities impossible.

In this paper, I depart from Kiefer's (2007) analysis and, following Giannakidou & Mari (2021), I assume that (i) ability is a modal category and (ii) that dispositionality and circumstantiality are constituent parts of ability.

- (27) *Anna ki tud-t-a rak-ni a Rubik-kockát.*
 Anna PRT be_able_to-PST-3SG put-INF the Rubik.cube.ACC
 'Anna could solve the Rubik cube.'

That is, sentence (28) means that:

- (28) In every accessible possible world where it is the case that:
- 1) John possesses the general ability of solving the Rubik-cube and
 - 2) John's internal conditions allow him to solve the Rubik-cube (dispositionality)
 - 3) John's external conditions allow him to solve the Rubik-cube (circumstantiality)
 - 4) John wants to solve the Rubik-cube;
- it is also the case that John actually solves the Rubik-cube.

Because of this, it is unwarranted to ascribe a separate circumstantial modality to the root infinitives under discussion. Rather, these sentences exhibit ability modality, of which circumstantiality (i.e., the limiting role of external circumstances) is an inherent component.¹⁰

4. Dispositional middles

In dispositional middles, the external argument of a transitive verb is syntactically suppressed and the sentence is interpreted as generic ability modality statement.¹¹ Strikingly, the verbs that participate in this construction are a proper subset of the ones that cannot combine with the ability modal *tud* 'be_able_to' (cf. also Appendix 1 and 2):

- (29) a. *Innen lát-sz-ik a hegycsúcs.*
 here.ELA BE_ABLE_TO.see-MID-3SG the summit
 'One can see the summit from here.'
- b. *Miről ismer-sz-ik meg egy jó cipő?*
 what.DEL BE_ABLE_TO.recognize-MID-3SG PRT a good shoe
 'How can one recognize a good pair of shoes?'

Verbs of involuntary perception and (re)cognition all currently have or historically used to have a dispositional middle form (with a few exceptions that can be reasonably characterized as accidental).¹² Verbs of physical disposition do not have a dispositional middle form. Some of them

¹⁰ This multi-component analysis of ability explains the following situation. Imagine that John is standing on his balcony with a view of the ocean, however, because of bad weather, the ocean is not visible. In that case, it is natural to say *John cannot see the ocean from his balcony*. even if under favourable conditions, the ocean is visible from the balcony. This is because in the sentence quote, what we negate is not the general ability in itself, but rather, the conjunction of general ability, disposition and circumstance.

¹¹ Cf. Halm (2020) for a comprehensive discussion of middle voice in Hungarian; a broad phenomenon which, in addition to dispositional middles, also includes anticausatives, mediopassives, reflexives, reciprocals and antipassives.

¹² These exceptions are examined in detail in Appendix 1. Middle voice paradigms have been gradually losing their productivity since the Middle Hungarian period, and in Modern Hungarian, mediality is at most a semi-productive phenomenon (Halm 2020). Because of this, the different paradigms of middle voice (including dispositional middles)

are intransitive (and thus, excluded from the middle paradigm), such as $\{be|k\acute{e}|át|el\}f\acute{e}r$ ‘BE_ABLE_TO.fit into, through, in’. However, transitive verbs of physical disposition (such as *elér* ‘BE_ABLE_TO.reach’) also do not have a dispositional middle form. This is plausibly a thematic restriction: middles need internal arguments that are either patients or stimuli/percepts. Since verbs of physical disposition have theme internal arguments, they fail to qualify.

To sum up, this means that having lexically specified ability modality is a necessary but insufficient condition for appearing in a dispositional middles. Two additional conditions apply: the verb has to be transitive and has to have a patient or stimulus/percept internal argument. Both of these additional conditions, however, are general constraint of the middle paradigm, and thus, they are orthogonal to the concerns of this paper.

Following Alexiadou & Doron (2002), Halm (2020) assumed that the source of ability modality in dispositional middles is a silent modal operator high in the syntax. Under my current proposal, there is no need for such stipulation as the modality is lexically hard-wired. This also explains why dispositional middles are limited to exactly this semantically well-defined and closed class of verbs in Hungarian.

Assuming for simplicity that GEN stands for universal quantification over relevant entities, the semantics of (29a) is the following (cf. Carlson 1989):

- (30) $\emptyset_{GEN}$ *lát-sz-ik* *a* *begyűűcs.*
 GEN BE_ABLE.see-MID-3SG the summit
 $GEN[\text{person}(x); \forall w' \in W [R^x_{\text{ability}}(w)(w') \rightarrow \text{see}^{w'}(ty.\text{summit}(y), x)]]$
 ‘One can see the summit. / The summit is visible. (Generally speaking, it is the case that all people can see the summit from here.)’

(30) can be paraphrased as follows: it is generally true of every (human) person that in every possible world w' which is compatible with their abilities in w and where they are cognizant of their ability and choose to exercise it, they actually see the summit.

It is well known that in many languages (such as English or Hebrew), the range of verbs participating in dispositional middles is wider and more diverse. In such languages, it remains plausible to assume that the source of modality is non-lexical: very possibly, a silent modal operator introduced high in the syntax. This is all the more plausible in languages such as English, where verbs of involuntary perception, (re)cognition and physical disposition can freely combine with an ability modal auxiliary, which clearly shows that ability modality is not part of their lexical semantic representation.

Above, following Halm (2020), I have assumed the the middle suffix *-sz-* (MID) is responsible for the syntactic suppression of the external argument, whereas genericity is introduced by a higher silent operator (GEN).¹³ There is, however, one striking empirical pattern that warrants a reconsideration of this position. While in Hungarian, there is in general a highly fragmented syncretism between various flavours of middle voice (anticausative, medio-passive, antipassive etc.) and the various forms of the MID suffix (*-(V)k\acute{e}V\acute{e}cs-*, *-Vd-* stb.)¹⁴; when it comes dispositional middles and the suffix *-sz-*, there is a strong, biconditional relationship: *-sz-* MID expones dispositional middles only, and all dispositional middles have *-sz-* MID.

This pattern can in principle be modelled in two ways. One may assume that the suffix *-sz-* expones both syntactic argument suppression and genericity (the GEN operator) as a portmanteau

exhibit numerous ad hoc lexical exceptions and gaps. However, what remains constant all over the time is that belonging to the group of verbs that are incompatible with *tud* ‘be able to’ is a necessary condition of appearing in dispositional middles. As middle voice loses its productivity, more and more accidental gaps appear in the paradigm: this process, however, is orthogonal to the concerns of this paper.

¹³ Following Alexiadou & Doron (2002), I assume that in dispositional middles, the syntactic slot of the external argument (Spec,VoiceP) is not projected. Because of this, the external argument position of the predicate is left unsaturated, i.e., it remains filled by a free variable which is then bound by the generic operator.

¹⁴ For a more detailed discussion, cf. Halm (2020: 30)

morpheme: in this case, the challenge is to account for the fact that the generic operator takes scope from a position much higher than *-sz-*. Alternatively, one may assume that there is indeed a silent generic operator high in the syntax, and it is this operator which licenses the appearance of the *-sz-* allomorph of the MID suffix (contextual allomorphy, cf. Bobaljik–Harley 2017). In this latter approach, *-sz-* only encodes argument suppression as a purely middle suffix, while the allomorphy is conditioned by the presence of the GEN operator. As the choice between these two accounts is orthogonal to the concerns of the present paper, we leave further discussion to later work.

5. As ability modal auxiliaries

Some verbs of involuntary perception, (re)cognition and physical disposition (*lát* ‘see’, *ért* ‘understand’, *bír* ‘endure’) can function as ability modal auxiliaries themselves (see also Appendix 1 and 2):¹⁵

- (31) a. *Félig-meddig megvakul-t-am, nem lát-ok olvas-ni.*
 sort.of PRT-blind_v-PST-1SG not be_able_to_{visual}-1SG read-INF
 ‘I am half blind, I can’t read (due to a degradation of my ability to see).’
 b. *Földrajzból nem ért-em meg-csinál-ni*
 geography.ELA not be_able_to_{understanding}-1SG PRT-do-INF
a házi-t, kérhetek segítséget?
 the homework.ACC ask.POSS.1SG help.ACC
 c. *Másfél óra után már nem bír-ok*
 1.5 hour after already not be_able_to_{endurance}-1SG
figyel-ni az előadásra.
 pay.attention.to-INF the talk.SUB
 After one and a half hours, I can no longer pay attention to the talk (due to my lack of mental or physical endurance/stamina).

The dialectal spread and level of acceptability of these auxiliary uses varies considerably. (See Appendix 2 full further data.):

- (32) a. *bír*
 i. BE_ABLE_TO.endure verb of disposition
 ii. be_able_to_{endurance} ability modal (fully acceptable)
 b. *tud*
 i. BE_ABLE_TO.know verb of cognition
 ii. be_able_to_{knowledge}¹⁶ ability modal (fully acceptable)
 c. *győz*
 i. BE_ABLE_TO.cope_with verb of disposition
 ii. be_able_to_{capacity} ability modal (limited to formal registers)
 d. *talál*
 i. BE_ABLE_TO.find verb of (re)cognition
 ii. be_able_to_{accidental} ability modal (limited to formal registers)
 e. *lát*

¹⁵ The modal auxiliary usage of *bír* ‘be_able_to_{endurance}’, *lát* ‘be_able_to_{visual}’, *talál* ‘be_able_to_{accidental}’ and *tud* ‘savoir; be_able_to_{knowledge}’ (note that this is distinct from the general ability modal *tud* ‘pouvoir; be able to’) is discussed by Kálmán et al. (1989).

¹⁶ This is the so-called *savoir* reading of *tud* (cf. Appendix 2). To sum up, *tud* is three-way ambiguous in Hungarian:

- (i) *tud*₁ ‘know’ main verb, verb of cognition
 (ii) *tud*₂ ‘savoir, be_able_to_{knowledge}’ ability modal (restricted to cognitive ability)
 (iii) *tud*₃ ‘pouvoir, be_able_to’ ability modal (non-restricted)

- | | | |
|-----|---------------------------------------|---|
| i. | BE_ABLE_TO.see | verb of perception |
| ii. | be_able_to _{visual} | ability modal (acceptable to most speakers) |
| f. | <i>ér-kező-ik</i> | |
| i. | BE_ABLE_TO.reach.MID | verb of disposition |
| ii. | be_able_to _{scheduling} -MID | ability modal (dialectal/archaic) |
| g. | <i>ér</i> | |
| i. | BE_ABLE_TO.reach | verb of disposition |
| ii. | be_able_to _{scheduling} | ability modal (dialectal/archaic) |
| h. | <i>ért</i> | |
| i. | BE_ABLE_TO.understand | verb of cognition |
| ii. | be_able_to _{understanding} | ability modal (very marginally acceptable) |
| i. | <i>lel</i> | |
| i. | BE_ABLE_TO.find | verb of (re)cognition |
| ii. | be_able_to _{accidental} | ability modal (very marginally acceptable) |

With the exception of *tud* ‘be_able_to_{knowledge}’ and *talál/lel* ‘be_able_to_{accidental}’, the acceptability of auxiliary use improves considerably if the verbs are in the scope of a negation operator:

- (33) a. *?**János vissza bír-ja tart-ani a kíváncsiságát.*
 John PRT be_able_to_{endurance}-3SG hold-INF the curiosity.3SG.ACC
 ‘John can hold back his curiosity.’
- b. *János nem bír-ja visszartart-ani a kíváncsiságát.*
 John not be_able_to_{endurance}-3SG PRT.hold-INF the curiosity.3SG.ACC
 ‘John can’t hold back his curiosity.’
- c. *János nehezen bír-ja visszartart-ani a kíváncsiságát.*
 John with.difficulty be_able_to_{endurance}-3SG PRT.hold-INF the curiosity.3SG.ACC
 ‘It is difficult for John to hold back his curiosity.’

No such contrast can be observed in the case of non-auxiliary use:

- (34) a. *János bír-ja a hideget.*
 John BE_ABLE_TO.endure-3SG the cold.ACC
 ‘John can take cold weather well.’
- b. *János nem bír-ja a hideget.*
 John not BE_ABLE_TO.endure-3SG the cold.ACC
 ‘John can’t tolerate cold weather.’
- c. *János nehezen bír-ja a hideget.*
 John with.difficulty BE_ABLE_TO.endure-3SG the cold.ACC
 ‘It is difficult for John to tolerate cold weather.’

The explanation of this pattern is left for later work. Regarding the phenomenon itself, my claim is that these modal auxiliary usages are the result of a grammaticalization process. Note that starting point of the reinterpretation is always a main verb which has lexically specified ability modality (35a). Because of this, the grammaticalization process can be analyzed as a kind of type shift, where the main verb with lexically hard-wired ability modality is reinterpreted as an ability modal, and the type of ability (visual, cognitive) etc. is re-encoded in the restrictor of the modal (35b).

- (35) a. $[[\text{lát}_{\text{main verb}}]]^{\text{w,g}} = \lambda x. \lambda y. \forall w' \in W [R^y_{\text{ability}}(w)(w') \rightarrow \text{see}^{w'}(y,x)]$

$$b. [[\textit{lát}_{\text{modal auxiliary}}]]^{w,g} = \lambda q. \forall w' \in W [R^y_{\text{visual ability}}(w)(w') \rightarrow q(w')]$$

(35b) is interpreted as follows: the semantic value of the modal auxiliary *lát* is a function which assigns the value TRUE to a proposition if and only if for all possible worlds w' that are compatible with the visual abilities in world w of the denotation of the first argument of the proposition, it is the case that, if the denotation of the first argument of the proposition is aware of their abilities and chooses to exercise them, then the proposition is to (in w'). Concretely, *Mari lát olvasni.* is true iff in every possible world that is compatible with our current world with regard to Mary's visual abilities, it is the case that if Mary is aware of her visual abilities and choosed to exercise them, she reads things.

The main claim of my paper offers a plausible explanation as to why it is exactly these verbs that have been open to such reanalysis. Since these verbs already inherently, lexically contained the element of ability modality, it was easy to reinterpret them by simply refactoring their non-modal component into the restrictor of a bona fide modal auxiliary. This can be regarded as a special instance of the general tendency of semantic bleaching in the course of grammaticalization. Additionally, in this process, a lexical element gets reinterpreted as a functional element: another well-known hallmark of grammaticalization. (Cf. Heine & Kuteva 2002; Gelderen 2004 a.o.). Finally, grammaticalization is known to typically involve reinterpretation as an element of higher semantic type (Von Stechow 1995), which is also the case here.

6. FCI-licensing

So-called free-choice items (FCIs) need to be licensed by a modal (or generic) operator in Hungarian (cf. Halm 2016a,b):

- (36) a. *#Tegnap este bárki megnézte a naplementét.*
 yesterday evening anyone PRT.watch.PST.3SG the sunset.ACC
 intended: 'Yesterday evening, anyone watched the sunset.' (non-modal)
- b. *Tegnap este bárki megnézhet-te a naplementét.*
 yesterday evening anyone PRT.watch-POSS-PST.3SG
 the sunset.ACC
 'Yesterday evening, anyone could watch the sunset.' (possibility)
- c. *Tegnap este bárki meg tudta nézni a naplementét.*
 yesterday evening anyone PRT. be_able_to.pst.3sg watch-INF
 the sunset.ACC
 'Yesterday evening, anyone was able to watch the sunset.' (ability)

A possibility modal suffix (36b) or an ability modal auxiliary (36c) can have scope over and thus license an FCI.¹⁷ Our claim is that when it comes to inherently modal verbs, the modality is specified extremely low, at the lexical level – and thus, we expect its scope to be very narrow. Thus, our prediction is that verbs with lexically specified ability modality cannot take scope above FCIs, and as a consequence, cannot license them. This is borne out empirically:

- (37) *#Tegnap este bárki lát-t-a*

¹⁷ Halm (in preparation) argues, based on the examination of complex scope patterns, that the modal operator takes scope from the head position of a ModP projection adjoined to the extended VP. Since the morphosyntactic exponent of the modal operator may be a bound suffix (*-hat*), Halm (in preparation) assumes that the operator is raised to this scope-taking position via feature movement. It is plausible to assume that a lexically hard-wired modal operator is not accessible to feature movement.

yesterday evening anyone BE_ABLE_TO.see-PST-3SG
a naplementét.
 the sunset.ACC
 intended: ‘Yesterday evening, anyone was able to see the sunset.’

In contrast to (37) above, (38) below is perfectly acceptable: however, here, it is modal operator expounded by the suffix *-hat* which take scope over and licenses the FCI:

- (38) *Tegnap este bárki lát-hat-t-a*
 yesterday evening anyone BE_ABLE_TO.see-POSS-PST-3SG
a naplementét.
 the sunset.ACC
 ASSERTION: ‘yesterday evening, it was possible concerning anyone that they had the ability to see the sunset.’
 ENTAILMENT: ‘yesterday evening, it was possible concerning anyone that they saw the sunset’

It is important to note that FCIs are licensed both in root infinitivals of circumstantial modality (39a) and in sentences containing a dispositional middle (39b):

- (39) a. *Innen lát-ni bármely-ik hegycsúcsot.*
 from.here BE_ABLE_TO.see-INF any-PART summit.ACC
 ‘From here, one can see any one of summits.’
 b. *Innen lát-sz-ik bármely-ik hegycsúcs.*
 from.here BE_ABLE_TO.see-MID-3SG any-PART summit
 ‘From here, any one of summits is visible.’

Since ability modality is encoded lexically here as well and supposedly has narrow scope, it is at first sight unclear what exactly licenses FCIs in these sentences. It should be noted, however, that both root infinitives and dispositional middles express generic statements, and FCIs are known to be licensed in generics in many languages, including Hungarian (Halm 2015; 2016):

- (40) a. *Ez a gyerek bármit meg-esz(-ik).*
 this the child anything.ACC PRT-eat-3SG
 ‘This child eats anything.’ (I.e., generally speaking, they are not picky when it comes to food.)
 GENERIC STATEMENT
 b. *#Ez a gyerek tegnap este fél 7 és 7 között bármit meg-ev-ett.*
 this the child yesterday evening half 7 and 7 between
 anything.ACC PRT-eat-PST.3SG
 intended ‘Yesterday evening, between half past six and seven, this child ate anything.’
 EPISODIC STATEMENT

Bartos (2002) argues convincingly for the presence of a generic operator in root infinitivals. Halm (2020) also posits a generic operator in sentences containing a dispositional middle. Thus, it appears straightforward that in the constructions, the FCI is licensed by a generic operator.

7. Why?

So far, we have seen that there is convincing, independent and converging evidence that verbs of involuntary perception, (re)cognition and physical disposition have an enriched, lexically specified ability modal semantics in Hungarian. This leads us to the following question: why is it exactly these verbs (and not others) that have lexically specified modality?

My proposal is that this phenomenon is driven by the general principle of economy. Syntactic operations are costly: therefore, if a particular verb is very often used in the scope of an ability modal operator, it might be more economical to encode this modality already in the lexicon, as opposed to introducing it in the syntax via an affix or an auxiliary.

This lexical solution does come with a potentially hefty cost, however: ability modality is hardwired into the semantics of these verbs: if we want to use the verb *lát* ‘see’ in a sentence, we are obliged to include the modal component too: *lát* ‘BE_ABLE_TO.see’. This creates two types of problems. What if we want to assert the actuality (*John sees the ocean.*) but not the ability (*John can see the ocean.*)? Technically, this is impossible: because of the lexical hard-wiring, in Hungarian, if we want to assert John’s seeing of the ocean, we can only do this by asserting John’s ability to see the ocean: we cannot say *John sees the ocean.* without saying that *John can see the ocean.* Note, however, that this is not really a problem: if someone actually sees the ocean, it follows that they possess the ability to see the ocean. So, while we are being redundant, we are not forced to assert anything that is not true.

The second problem is arguably more serious. Because of the lexical hard-wiring, we have no way of directly asserting the actuality (*John sees the ocean.*), we can only assert the ability (*John can see the ocean.*). For a very large class of verbs, this would be fatal: if *dance*, for example, had a similar lexically enriched semantics, we would only be able to assert ‘x is able to dance’; i.e., ‘x is dancing’ would be ineffable. This would drastically reduce the expressive power of language.

While in general, the second problem seems fatal, can we think of a verb class that is immune to it? The answer is yes: if a verb projects a predicate p , of which it is true that *able p* logically entails p , the ineffability problem ceases to be fatal. While the speaker wishing to assert p can only assert *able p*, they can trust the hearer to calculate the logical entailment to p , and to add both *able p* and p to their representation of the Common Ground, in line with the communicative intent of the speaker. Thus, it is no coincidence that in Hungarian, it is verbs of involuntary perception, (re)cognition and physical disposition that have hard-wired lexical ability modality: as we have seen, with these verbs, *able p* does indeed logically entail p .

At this point, however, a further question arises: if such a lexical strategy is available and economical, why do all languages not follow this strategy? It should be noted that while this strategy is indeed economical in some respects, it also comes with certain extra costs. On the economy side, the need to merge (and possibly move) a modal auxiliary or affix is eliminated. However, two costs also arise: (i) redundancy (the need to explicitly assert ability even if we only want to assert actuality) and (ii) the need to calculate logical entailment to arrive from *able p* to p (which necessitates extra effort on the side of the hearer.)

In sum, using the lexical strategy reduces the functional load on syntax, at the expense of increasing the functional load on the lexicon (ability modality is lexically specified) and on semantics (logical entailment needs to be calculated). Also, the lexical strategy changes the division of labour between speaker and hearer. The workload of the speaker is reduced (they do not have to merge and possibly move a modal auxiliary or affix in the syntax) and the workload of the hearer is increased (the inference from modality to actuality has to be calculated by the hearer). So it is not the case that the lexical strategy is globally superior to the syntactic strategy: rather, the two strategies instantiate two different patterns of division of labour between (i) the various modules of language and (ii) between speaker and hearer.

Throughout this paper, I assumed an architecture where the lexical entry of the verbs concerned contains an ability modal element. This is not without precedent (cf. the expanding literature on sublexical modality in English and other languages, starting with Koenig & Davis

2021, cf. Martin 2025 for recent overview.) This means that grammatical modality can enter the derivation in two places, so to speak: at the lexical level (via verbs having a lexically specified modal element) and also later, at the morphosyntactic level (via modal affixes or modal auxiliaries). One potential analytical alternative to this would be to assume that grammatical modality can only enter the derivation in the morphosyntax: in such a framework, one might argue that the incompatibility of the verbs under discussion with an overt ability modal *tud* ‘be able to’ is the result of a restriction to the effect that these verbs can only associate with a covert ability modal. Such an alternative approach would not be without appeal: in addition to having a simpler architecture, it would also offer an elegant treatment of the appearance of an overt ability modal in emphatic contexts. However, it would also make one strikingly wrong prediction: if the verbs under discussion indeed invariably appeared as complements of silent but syntactically active modals, we would expect them to surface in infinitival form, contrary to fact. Because of this, the ‘silent modal’ analysis is, on balance, much less attractive than our proposal.

8. Conclusion

The empirical starting point of our proposal was the following new set of observations: in Hungarian, verbs of involuntary perception, (re)cognition and physical disposition form a natural class on the basis of their syntactic behaviour across four, seemingly unrelated phenomena: (i) incompatibility with the ability modal *tud* ‘be able to’; (ii) availability in root infinitivals; (iii) availability in dispositional middles and (iv) ability to function as modal auxiliaries.

As I have argued above, this striking pattern receives a natural and convincing explanation under the assumption that verbs falling into these classes have an enriched lexical semantics: ability modality is part of their lexical semantic specification. I have also shown that it is no accident that this is exactly this set of verbs that have such an enriched semantics, with voluntariness being the key factor. Finally, I have argued that cross-linguistically, the lexical encoding of ability modality can be viewed as an optional design element in the architecture of language which modulates the division of labour between (i) the various modules of language (syntax and semantics) and (ii) between speaker and hearer.

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Appendix 1: Verbs and constructions – Tabulated overview

		incompatibility with <i>tud</i> 'be_able_to'	root infinitive of circumstantial modality	dispositional middle	as modal auxiliary
involuntary perception	<i>lát</i> 'be_able_to.see'	✓	✓	✓	✓
	<i>hall</i> 'be_able_to.hear'	✓	✓	✓	✗
	<i>érez</i> 'be_able_to.feel'	✓	✓	✓	✗
involuntary (re)cognition	<i>felismer</i> 'be_able_to.recognize'	✓	✓	†	✗
	<i>megismer</i> 'be_able_to.recognize'	✓	✓	†	✗
	<i>ismer</i> 'be_able_to.know'	✓	✓	†	✗
	<i>tud</i> 'be_able_to.know'	✓	✓	✗	✓
	<i>tapasztal</i> 'be_able_to.experience'	✓	✓	✗	✗
	<i>ért</i> 'be_able_to.understand'	✓	✓	%	%
	<i>emlékezik</i> 'be_able_to.remember'	✓	✓	(intransitive)	✗
	<i>talál</i> 'be_able_to.find'	✓	✓	%	✓
	<i>kap</i> 'be_able_to.find'	✓	✓	%	✗
	<i>lel</i> 'be_able_to.find'	✓	✓	†	%
physical disposition	<i>olvas</i> 'be_able_to.find_by reading'	✓	✓	✗	✗
	<i>elér</i> 'be_able_to.reach'	✓	✓	✗	%
	<i>bír</i> 'be_able_to.endure'	✓	✓	✗	✓
	<i> kibír</i> 'be_able_to.endure'	✓	✓	✗	✗
	<i>elbír</i> 'be_able_to.carry'	✓	✓	✗	✗
	<i>tűr</i> 'be_able_to.endure'	✓	?	✗	✗
	<i>győz</i> 'be_able_to.cope_with'	✓	✓	✗	✓
	<i>áll</i> 'be_able_to.withstand'	✓	?	✗	✗
	<i>befér</i> 'be_able_to.fit into'	✓	✓	(intransitive)	✗
	<i>átfér</i> 'be_able_to.fit through'	✓	✓	(intransitive)	✗
	<i>kiszér</i> 'be_able_to.fit through on way out'	✓	✓	(intransitive)	✗
	<i>elfér</i> 'be_able_to.fit inside'	✓	✓	(intransitive)	✗

Legend: ✓: yes
✗: no
†: archaic
%: dialectal
?: marginal

Table 1: Verbs and constructions

Table 1 reveals the following patterns. Verbs falling into the semantic classes under discussion (verbs of involuntary perception, involuntary (re)cognition and physical disposition) are characterized by lexically specified ability modality. The test of this property is the incompatibility with the ability modal *tud* 'be able to'. All verbs falling into these classes are incompatible with *tud*, and only the verbs falling into these classes are incompatible with *tud*.

Lexically specified ability modality is a necessary and sufficient condition of the appearance in root infinitivals of circumstantial modality.

Lexically specified ability modality is a necessary but insufficient condition of the appearance in dispositional middles. Additional conditions are: transitivity (trivially) and a percept/stimulus internal argument. Verbs of cognition appear to be partial exceptions to this generalization: while *ismer* 'BE_ABLE_TO.know' has a dispositional middle form (*ismer-sz-ik*), *tud* 'BE_ABLE_TO.know' and *tapasztal* 'BE_ABLE_TO.know' lack such a form (**tud-sz-ik*, **tapasztal-sz-ik*). These are probably accidental gaps in the at most semi-productive middle paradigm (which is known to have numerous accidental gaps, cf. Halm 2020 and also discussion above).¹⁸

Whether a verb has been reinterpreted as a modal auxiliary in the course of grammaticalization appears to be largely accidental: *lát* 'BE_ABLE_TO.see' has been reinterpreted, whereas *hall* 'BE_ABLE_TO.hear' and *érez* 'BE_ABLE_TO.feel' have not; *tud* 'BE_ABLE_TO.know' and *talál* 'BE_ABLE_TO.find' have been reinterpreted, whereas *ismer* 'BE_ABLE_TO.know' and *lel* 'BE_ABLE_TO.find' not; *bír* 'BE_ABLE_TO.endure' and *győz* 'BE_ABLE_TO.cope_with' have been

¹⁸ The form (*ki*)-*tud-ód-ik* does exist, however, this is not a dispositional middle but an anticausative (or possibly medio-passive) form: while the external argument is syntactically suppressed here as well, genericity and modality are both absent. The form *tud-at-ik* was also available historically, however, this is not a middle but a passive form.

reinterpreted, whereas *tűr* ‘BE_ABLE_TO.endure’ and *áll* ‘BE_ABLE_TO.withstand’ not. It appears that comparing verbs with similar meaning, it is the more frequent ones that were prone to reinterpretation (figures in parentheses are the number of hits in the Hungarian National Corpus, verbs that have undergone reinterpretation are bolded):¹⁹

- (41) a. ***lát*** (1.019.119) > *érez* (181.204) > *hall* (20.501)
 b. ***tud*** (2.899.897) > *ismer* (431.780)
 c. ***talál*** (524.388) > *lel* (19.501)
 d. ***bír*** (135.443) > ***győz*** (95.988) > *tűr* (21.495)

Finally, we discuss two verbs which may appear to be outliers. Note first that the verbs *talál* ‘BE_ABLE_TO.find’ and *lel* ‘BE_ABLE_TO.find’ are clearly verbs of involuntary perception: while searching for something is voluntary, the actual finding of it is a punctual involuntary event, the core of which is the involuntary perception and recognition of the object being searched for.

Kap is polysemous between two meanings: *kap*₁ ‘receive, get’ and *kap*₂ ‘BE_ABLE_TO.find’. In the constructions under discussion here, it is clearly used in the second sense, as a synonym of *talál* ‘BE_ABLE_TO.find’:

- (42) *Nehezen kap-ni taxit Budapesten.*
 with.difficulty find-INF taxi.ACC Budapest.SUP
 ‘It is difficult to find a taxi in Budapest.’

This is not unexpected. It is well-known that in certain dialects of Hungarian (such as the Székler dialect) *kap*₂ is the most frequently used verb to describe an event of finding, and even in Standard Hungarian, such usage is well-attested:

- (43) *A Nyugati téren szinte lehetetlen
 the Nyugati square.SUP almost impossible
 parkolóhelyet kap-ni.
 parking.lot.ACC find-INF
 ‘It is almost impossible to find parking space at Nyugati square.’*

The other possible outlier is *olvas* ‘read’, which we categorized in Table 1 as a verb of involuntary (re)cognition. *Olvas* is well-attested in infinitivals of circumstantial modality (see Appendix 2):

- (44) *Mindenhol azt olvas-ni, hogy
 everywhere that.ACC read-INF that
 a terhesség-re nem veszélyes
 the pregnancy.SUB not dangerous
 a COVID-19 fertőzés.
 the COVID-19 infection
 ‘What you can read everywhere is that COVID-19 does not represent a risk to pregnancy.’
 (‘The information that you can find by reading everywhere is that COVID-19 does not represent a risk to pregnancy.’)*

However, in all these instance, *olvas* is interpreted with the meaning of ‘find something in a text by way of reading’. Dömötör (2025:63) reaches a similar conclusion after a detailed analysis of corpus

¹⁹ The hits reported include both main verb and auxiliary uses; however, since with most verbs, the main verb use is vastly more frequent, they give a good illustration of the relative frequencies; with the exception of *tud*, where the auxiliary use is also very frequent. I did not include the figures for *áll* (431.780), which is polysemous between *áll*₁ ‘stand’ and *áll*₂ ‘BE_ABLE_TO.withstand’.

data: „[az olvas] a [szóban forgó] példákban a 'könyvben, szövegben (meg)talál valamit' szinonimájaként értelmezhető”.²⁰

Appendix 2: Verbs and constructions – Sentence samples

In this appendix, I present sentence samples. Corpus data reported are drawn from the Hungarian National Corpus (Oravecz et al. 2014).

1. Incompatibility with the ability modal auxiliary *tud* ‘be able to’

1.1. Verbs of involuntary perception

lát ‘BE_ABLE_TO.see’

- (45) a. #*A vak újra tud lát-ni.* 0 HNC-hits²¹
the blind again be_able_to-3SG BE_ABLE_TO.see-INF
intended: ‘The blind man can see again.’
b. *A vak újra lát.* 1190 HNC-hits²²
the blind again BE_ABLE_TO.see-3SG
‘The blind man can see again.’

hall ‘BE_ABLE_TO.hear’

- (46) a. #*Tudsz hall-ani engem?*
be_able_to.2SG BE_ABLE_TO.hear-INF me?
intended: ‘Can you hear me?’
b. *Hallasz engem?*
BE_ABLE_TO.hear-2SG me
‘Can you hear me?’

Contrast this with a verb of voluntary perception such as *hallgat* ‘listen’:

- (47) a. *Tudsz zenét hallgat-ni a munkahelyeden?*
be_able_to-2SG music.ACC listen-INF the office.your.on
‘Can you listen to music at your office?’
b. *Hallgat-sz zenét a munkahelyeden?*
listen-2SG music.ACC the office.your.on
‘Do you listen to music at your office?’ -> only a non-ability, habitual reading is available

In (47b), only a non-modal, habitual reading is available. Consider also:

[Scenario: John received a new hearing implant.]

- (48) a. #*János újra tud hall-ani!* 0 HNC-hits²³
John again be_able_to-3SG BE_ABLE_TO.hear-INF

²⁰ In the examples discussed above, *olvas* can be interpreted as a synonym of ‘find something in a book or text’

²¹ CQL search expression: [lemma="újra"] [lemma="tud"] [word="látni"]

²² CQL = [lemma="újra"] [lemma="lát"]

²³ CQL = [lemma="újra"] [lemma="tud"] [word="hallani"]

- intended: 'John can hear again.'
- b. *János újra hall!* 8 HNC-hits²⁴
 John again BE_ABLE_TO.hear-3SG
 'John can hear again.'

érez BE_ABLE_TO.feel 'be able to feel'

[Scenario: a medical emergency]

- (49) a. *#Tud-od érez-ni a pulzusát?* 0 HNC-hits²⁵
 be_able_to-2SG BE_ABLE_TO.feel-INF the pulse.her.ACC
 intended: 'Can you feel her pulse?'
- b. *Érzed a pulzusát?* 20 HNC-hits²⁶
 BE_ABLE_TO.feel-2SG the pulse.her.ACC
 'Can you feel her pulse?'

Contrast this with *ki-tapint* OUT-touch 'locate/get to feel by voluntarily touching'

- (50) a. *Ki tud-od tapint-ani a pulzusát?*
 OUT be_able_to.2SG touch-INF the pulse.her.ACC
 'Can you find her pulse?'
- b. *Ki-tapint-od a pulzusát?*
 OUT- touch-2SG the pulse.her.ACC
 'Will you find her pulse?' (in Hungarian, present tense can have a future interpretation)

1.2. Verbs of involuntary (re)cognition

*meg-ismer*_i PRT-BE_ABLE_TO.recognize 'be able to recognize', *fel-ismer* UP-BE_ABLE_TO.recognize 'be able to recognize'

- (51) a. *#Nem tud-t-am {meg|fel}-ismer-ni*
 not be_able_to-PST-1SG PRT-BE_ABLE_TO.recognize-INF
Jánost, mert maszkban volt.
 John.ACC because mask.in was
 intended: 'I could not recognize John because he was wearing a mask.'
- b. *Nem ismer-t-em {meg|fel}*
 not BE_ABLE_TO.know-PST-1SG PRT
Jánost, mert maszkban volt.
 John.ACC because mask.in was
 'I could not recognize John because he was wearing a mask.'

Note that English *recognize* seems to have a similar, though much weaker, tendency:

- (52) a. *I could not recognize her voice.* 55.700 Google hits²⁷
 b. *I did not recognize her voice.* 345.00 Google hits

²⁴ CQL = [lemma="újra"] [lemma="hall"]

²⁵ CQL = [lemma="tud"] [word="érezni"] [word="a"] [lemma="pulzus | szívverés | lélegzés"]

²⁶ CQL = [lemma="érez"] [word="a"] [lemma="pulzus | szívverés | lélegzés"]

²⁷ A caveat: English *can* is ambiguous between various modality readings (ability, possibility, permission), unlike *tud* 'be.able.to'. This should be taken into account when interpreting the Google hit figures.

Contrast this to voluntary *meg-ismer*₂ PRT-know ‘get to know’:

- (53) a. *Sosem tud-t-am igazán jól meg-ismer-ni Marit.*
 never be_able_to-PST-1SG truly well PRT-know-INF Mary.ACC
 I could never get to know Mary really well.
 b. *Sosem ismer-t-em meg igazán jól Marit.*
 never know-PST-1SG PRT truly well Mary.ACC
 I did not ever get to know Mary really well.

emlékezik BE_ABLE_TO.remember ‘be able to remember’

- (54) a. *#Mari tud-ott emlékez-ni*
 Mary be_able_to-PST-3SG BE_ABLE_TO.remember-INF
János telefonszámára
 John number.3SG.SUB
 intended: ‘Mary could remember John’s phone number.’
 b. *Mari emlékez-ett János telefonszámára.*
 Mary BE_ABLE_TO.remember-PST-3SG John number.3SG.SUB
 ‘Mary could remember John’s phone number.’

Contrast this with voluntary *fel-idéző* UP-recall ‘recall’:

- (55) a. *Mari fel tud-ta idéző-ni*
 Mary UP be_able_to-PST-3SG recall-INF
a gyerekkori élményeit.
 the childhood experiences.her.ACC
 ‘Mary could recall her childhood experiences.’
 b. *Mari fel-idéző-te a gyerekkori élményeit.*
 Mary UP-recall-PST-3SG the childhood experiences.her.ACC
 ‘Mary recalled her childhood experiences.’

Contrast this with English *remember*:

- (56) a. *I did not remember his name.* 505.000 Google hits.
I could not remember her name. 1.811.000 Google hits.

ért BE_ABLE_TO.understand ‘be able to understand’

- (57) a. *#Nem tud-om ért-eni*
 not be_able_to-1SG BE_ABLE_TO.understand-INF
a formális szemantikát.
 the formal semantics.ACC
 intended: ‘I can’t understand formal semantics.’
 b. *Nem ért-em a formális szemantikát.*
 not BE_ABLE_TO.understand-1SG the formal semantics.ACC
 ‘I can’t understand formal semantics.’

Contrast this to voluntary *értelmez* ‘interpret, develop an understanding of’:

- (58) *Nem tud-t-am értelmez-ni a helyzetet.*
 not be_able_to-PST-1SG interpret-INF the situation.ACC

‘I could not develop an understanding of the situation.’

ismer BE_ABLE_TO.know ‘be able to know’

- (59) a. #*Nem tud-t-am jól ismer-ni őt*
 not be_able_to-PST-1SG well BE_ABLE_TO.know-INF her
azokban az években.
 those.in the years.in
 intended: ‘I was unable to know her very well in those years.’
 b. *Nem ismer-t-em őt jól*
 not BE_ABLE_TO.know-PST-1SG her well
azokban az években.
 those.in the years.in
 ‘I did not know her well in those years.’

As expected, verbs of voluntarily gaining cognition can freely combine with an ability modal:

- (60) *Nem tudtam jól meg-ismer-ni őt*
 not be_able_to-PST-1SG well PRT-get.to.know-INF her
azokban az években
 those.in the years.in
 ‘I could not get to know her well in those years.’

Indeed, while somewhat marked (probably due to a haplology violation), even *tud*₁ ‘BE_ABLE_TO’ and *meg.tud*₂ ‘get.to.know’ can combine fairly felicitously:

- (61) *Na le-ír-t-am mindent,*
 well DOWN-write-PST-1SG all.ACC
amit eddig meg tud-t-am tud-ni.
 which.ACC so.far PRT be_able_to-PST-1SG get.to.know-INF
 ‘I jotted down everything here that I was able to find out for the time being.’

talál BE_ABLE_TO.find ‘be able to find’

- (62) a. #*Nem tud-t-am parkolóhelyet talál-ni.* 0 HNC hits²⁸
 not be_able_to-PST-1SG parking.space.ACC BE_ABLE_TO.find-INF
 intended: ‘I could not find a parking space.’
 b. *Nem talál-t-am parkolóhelyet.* 20 HNC hits²⁹
 not BE_ABLE_TO.find-PST-1SG parking.space.ACC
 ‘I could not find a parking space.’

kap BE_ABLE_TO.find ‘be able to find’

- (63) a. #*Tud-t-am kap-ni parkolóhelyet.* 0 HNC hits³⁰
 be_able_to-PST-1SG BE_ABLE_TO.find-INF parking.space.ACC
 intended: ‘I could find a parking place.’
 b. *Kap-t-am parkolóhelyet.* 4 HNC hits³¹

²⁸ CQL = [word="nem"] [lemma="tud"] [lemma="parkoló|parkolóhely"] [word="találni"] / [word="nem"] [lemma="tud"] [word="találni"] [lemma="parkoló|parkolóhely"]

²⁹ CQL = [word="nem"] [lemma="talál"] [lemma="parkoló|parkolóhely"]

³⁰ CQL = [lemma="tud"] [word="kapni"] [lemma="parkoló|parkolóhely"]

³¹ CQL = [lemma="kap"] [lemma="parkoló|parkolóhely"]

BE_ABLE_TO.find-PST-1SG parking.space.ACC
 'I could find a parking space.'

lel BE_ABLE_TO.find 'be able to find'

- (64) a. #*Nem tud-ott nyugalmat lel-ni.* 1 HNC hit³²
 not be_able_to-PST-3SG calm.ACC BE_ABLE_TO.find-INF
 intended: 'She could not find peace.'
 b. *Nem lel-t nyugalmat.* 71 HNC hits³³
 not BE_ABLE_TO.find-PST-3SG calm.ACC
 'She could not find peace.'

Contrast this with English:

- (65) a. *She could not find any peace.* 11.770 Google hits
 b. *She did not find any peace.* 11 Google hits

1.3. Fizikaidiszpozíció-igék

elér PRT-BE_ABLE_TO.reach 'be able to reach'³⁴

- (66) a. #*Nem tudom el-ér-ni*
 not be_able_to-1SG PRT-BE_ABLE_TO.reach-INF
 a távirányítót.
 the remote.ACC
 intended 'The remote is out of my reach.'
 b. *Nem ér-em el a távirányítót.*
 not BE_ABLE_TO.reach-1SG PRT the remote.ACC
 'The remote is out of my reach.'

{*ki*} *bír* {PRT} -BE_ABLE_TO.endure 'be able to endure / cope with'

- (67) a. #*Mari tud-ja bír-ni* 0 HNC hits³⁵
 Mary be_able_to-3SG BE_ABLE_TO.endure-INF
 a kritikát.
 the criticism.ACC
 intended: 'Mary can take criticism.'
 b. *Mari bír-ja a kritikát.* 107 HNC hits³⁶
 Mary BE_ABLE_TO.endure-3SG the criticism.ACC
 'Mary can take criticism.'

³² CQL = [lemma="tud"] [lemma="nyugalom"] [word="lelni"] / [lemma="tud"] [word="lelni"] [lemma="nyugalom"]

³³ CQL = [lemma="lel"] [lemma="nyugalom"]

³⁴ Note that *el-ér* is polysemous. *el-érz* 'achieve + propositional argument' is a volitional verb and as such, can freely combine with *tud* 'be.able.to':

Mari elérte, hogy megemeljék a fizetését.

Mari arranged a pay rise for herself.

Mari el tudta érni, hogy megemeljék a fizetését.

Mary could arrange a pay rise for herself.

³⁵ CQL = [lemma="tud"] [word="bírn"] [word="a"] [lemma="kritika"]

³⁶ CQL = [lemma="bír"] [word="a"] [lemma="kritika"]

- (68) a. #*János nem tud-ja bír-ni* 0 HNC hits³⁷
 John not be_able_to-3SG BE_ABLE_TO.endure-INF
a hideget.
 the cold.ACC
 intended: 'John can't take the cold.'
- b. *János nem bírja a hideget.* 269 HNC hits³⁸
 John not BE_ABLE_TO.endure-3SG the cold.ACC
 'John can't take the cold.'

Notice the contrast with volitional *elvisel* 'put up with':

- (69) a. *János nem tud-ja el-visel-ni* 12 HNC hits³⁹
 John not be_able_to-3SG PRT-bear-INF
a kritikát.
 the criticism.ACC
 'John does not put up with any criticism.' 8 HNC hits⁴⁰
- b. *János nem viseli el a kritikát.*
 John not bear-3SG PRT the criticism.ACC
 'John can't put up with any criticism.'

That *el-visel* 'put up with' is volitional and {*ki*|}*bír* 'be able to endure / cope with' is not is illustrated below:

- (70) a. *Nem vagyok hajlandó el-visel-ni a mellőzést.*
 not am willing PRT-bear-INF the sidelining.ACC
 I am unwilling to put up with being sidelined.
- b. #*Nem vagyok hajlandó bír-ni a mellőzést.*
 not am willing BE_ABLE_TO.endure-INF
 the sidelining.ACC
 intended: 'I am unwilling to endure being sidelined.'

tűr BE_ABLE_TO.endure 'be able to endure/tolerate'

- (71) a. #*Jól tud-om tűr-ni a hideget.* 0 HNC hits⁴¹
 well be_able_to-1SG BE_ABLE_TO.endure-INF the cold.ACC
 intended: 'I can take the cold pretty well.'
- b. *Jól tűröm a hideget.* 18 HNC-hits⁴²
 well BE_ABLE_TO.endure-1SG the cold.ACC
 'I can take the cold pretty well.'

áll BE_ABLE_TO.withstand 'be able to withstand'

³⁷ CQL = [lemma="tud"] [word="bírn"] [word="a"] [lemma="hideg|meleg"]

³⁸ CQL = [lemma="bír"] [word="a"] [lemma="hideg|meleg"]

³⁹ CQL = [word="nem"] [lemma="tud"] [word="elviselni"] [word="a"]
 [lemma="kritika|kudarc|megaláztatás|szégyen"]

⁴⁰ CQL = [word="nem"] [lemma="visel"] [word="el"] [word="a"] [lemma="kritika|kudarc|megaláztatás|szégyen"]

⁴¹ CQL = [word="jól"] [lemma="tud"] [word="tűrn"] [word="a"] [lemma="hideg"]

⁴² CQL = [word="jól"] [lemma="tűr"] [word="a"] [lemma="hideg"]

- (72) a. #*János tud-ja áll-ni* 1 HNC hit⁴³
 John be_able_to-3SG BE_ABLE_TO.withstand-INF
az ütéseket.
 the hits.ACC
 intended: 'John can take the hits.'
- b. *János áll-ja az ütéseket.* 40 HNC hits⁴⁴
 John BE_ABLE_TO.withstand-3SG the hits.ACC
 'John can take the hits.'

el-bír PRT-BE_ABLE_TO.sustain 'be able to carry (something heavy)'

- (73) a. #*Mari el tud-ja bír-ni* 0 HNC hits⁴⁵
 Mary PRT be_able_to-3SG BE_ABLE_TO.sustain-INF
a bőröndöt.
 the suitcase.ACC
 intended: 'Mary can carry the suitcase.' 21 HNC hits⁴⁶
- b. *Mari el-bír-ja a bőröndöt.*
 Mary PRT-BE_ABLE_TO.sustain-3SG the suitcase.ACC
 'Mary can carry the suitcase.'

győz BE_ABLE_TO.cope_with 'be able to cope with'

- (74) a. #*Alig tud-ta győz-ni* 0 HNC hits⁴⁷
 barely be_able_to-PST-3SG BE_ABLE_TO.cope_with-INF
a sok munkát.
 the much work.ACC
- b. *Alig győz-te* 106 HNC hits⁴⁸
 barely be_able_to.cope_with-PST-3SG
a sok munkát.
 the much work.ACC

befér PRT-BE_ABLE_TO.fit 'be able to fit {into|through}'

- (75) a. #*Nem tud-t-am be-fér-ni* 0 HNC hits⁴⁹
 not be_able_to-PST-3SG IN-BE_ABLE_TO.fit-INF
az ajtón.
 the door.on
 intended: 'I could not fit through the door.'
- b. *Nem fér-t-em be az ajtón.* 29 HNC hits⁵⁰
 not BE_ABLE_TO.fit-PST-3SG PRT the door.on
 'I could not fit through the door.'

⁴³ CQL = [lemma="tud"] [word="állni"] [word="az"] [word="ütéseket"]

⁴⁴ CQL = [lemma="áll"] [word="az"] [word="ütéseket"]

⁴⁵ CQL = [word="el"] [lemma="tud"] [word="bírn"] [word="a | az"]
 [lemma="súly | teher | terhelés | kapa | eke | kistáska | táska | bőrönd"]

⁴⁶ CQL = [lemma="elbír"] [word="a | az"] [lemma="súly | teher | terhelés | kapa | eke | kistáska | táska | bőrönd"]

⁴⁷ CQL = [lemma="tud"] [word="győzni"] [word="a | az"]
 [word="munkát | harcot | loholást | küzdelmet | versenyt | iramot | versengést"]

⁴⁸ CQL = [lemma="győz"] [word="a | az"]
 [word="munkát | harcot | loholást | küzdelmet | versenyt | iramot | versengést"]

⁴⁹ CQL = [word="nem"] [lemma="tud"] [word="beférni"] [word="a | az"] [msd="FN.SUP"]

⁵⁰ CQL = [word="nem"] [lemma="fér"] [word="be"] [word="a | az"] [msd="FN.SUP"]

Contrast this to volitional *be-megy* PRT-go ‘go through’:

- (76) a. *Nem tud-t-am be-men-ni az ajtón.* 3 HNC hits⁵¹
 not be_able_to-PST-3SG PRT-go-INF the door.on
 ‘I could not go through the door.’
 b. *Nem men-t-em be az ajtón.* 3 HNC hits⁵²
 not go-PST-3SG PRT the door.on
 I did not go through the door.

Consider also:

- (77) a. *#János be tud-ott fér-ni* 0 HNC hits⁵³
 John IN be_able_to-PST-3SG BE_ABLE_TO.fit-INF
 a csapatba.
 the team.into
 intended: ‘John could make it to the team.’
 b. *János be-fér-t a csapatba.* 41 HNC hits⁵⁴
 John IN-be_able_to.fit-PST-3SG the team.into
 ‘John could make it to the team.’

Contrast this to volitional *be-illeszked-ik* PRT-position_v-MID-3SG ‘find.one’s place somewhere’:

- (78) a. *János be tud-ott illeszked-ni* 11 HNC hits⁵⁵
 John PRT be_able_to-PST-3SG position_v-MID-INF
 a csapatba.
 the team.into
 ‘John could find his place in the team.’
 b. *János be-illeszked-ett a csapatba.* 124 HNC hits⁵⁶
 John PRT-position_v-MID-PST-3SG the team.into
 ‘John found his place in the team.’

2. Root infinitivals of circumstantial modality

Every sentence presented in this section is an actual attested sentence, either in the Hungarian National Corpus or in Hungarian-language electronic texts available on the internet.

2.1. Verbs of involuntary perception

- (79) *Hall-ani a tenger morajlását.*
 BE_ABLE_TO.hear-INF the sea roar.its.ACC
 ‘One can hear the roar of the sea.’

⁵¹ CQL = [word="nem"] [lemma="tud"] [word="bemenni"] [word="a|az"] [msd="FN.SUP"]

⁵² CQL = [word="nem"] [lemma="megy"] [word="be"] [word="a|az"] [msd="FN.SUP"]

⁵³ CQL = [word="be"] [lemma="tud"] [word="féni"] [word="a|az"]
 [word="csapatba|keretbe|névsorba|delegációba|zenekarba|kórusba|válogatottba"]

⁵⁴ CQL = [lemma="befér"] [word="a|az"]
 [word="csapatba|keretbe|névsorba|delegációba|zenekarba|kórusba|válogatottba"]

⁵⁵ CQL = [word="be"] [lemma="tud"] [word="illeszkedni"] [word="a|az"]
 [word="csapatba|keretbe|névsorba|delegációba|zenekarba|kórusba|válogatottba"]

⁵⁶ CQL = [lemma="beilleszkedik"] [word="a|az"]
 [word="csapatba|keretbe|névsorba|delegációba|zenekarba|kórusba|válogatottba"]

- (80) *Innen lát-ni a hegycsúcsot.*
 from.here BE_ABLE_TO.see-INF the summit.ACC
 ‘One can see the summit from here.’
- (81) *Érez-ni a virágok illatát.*
 BE_ABLE_TO.feel-INF the flowers scent.their.ACC
 ‘One can feel the scent of the flowers.’

2.2. Verbs of involuntary (re)cognition

- (82) *Falusi Mariann angyali hangját ezer közül is meg-ismer-ni.*⁵⁷
 FalusiMariann angelic voice.her.ACC 1000 from too
 PRT-BE_ABLE_TO.recognize-INF
 ‘One can recognize the angelic voice of Mariann Falusi from among thousand others.’
- (83) *A ceglédi üzletkötőt a nyelhpiercing miatt kissé nehezen ért-eni.*
 the Cegléd salesman.ACC the tongue.piercing because.of
 a.bit hardly BE_ABLE_TO.understand-INF
 ‘The salesman from Cegléd is somewhat difficult to understand because of his tongue piercing.’
- (84) *Jellegzetes rajztechnikájáról könnyen fel-ismer-ni Jazava alkotásait.*
 typical drawing.technique.her.from easily
 PRT-BE_ABLE_TO.recognize-INF Jazava creations.her.ACC
 ‘One can easily recognize Jazava’s work because of her characteristic drawing technique.’
- (85) *Nem emlékez-ni konstruktív javaslatukra.*
 not BE_ABLE_TO.remember-INF constructive proposal.theirs.onto
 ‘One cannot remember any constructive proposal of theirs.’
- (86) *Az ütközések következtébengyakran tapasztal-ni motorhibát.*
 the collision.PL because.of often experience-INF motor.damage.ACC
 ‘Because of the collisions, one can often experience motor damage.’
- (87) *Manapság már ritkán talál-ni valódi antik szőnyeget.*
 these.days already rarely BE_ABLE_TO.find-INF
 authentic antique rug.ACC
 ‘These days one can rarely find an authentic antique rug.’
- (88) *Nehezen kap-ni taxit Zomborban.*⁵⁸
 with.difficulty BE_ABLE_TO.find-INF taxi.ACC Zombor.in
 ‘One can only find a taxi in Zombor with difficulty.’
- (89) *Nem lel-ni a nyomát Orbán kutyájának.*
 not BE_ABLE_TO.find-INF the trace.its.ACCOrbán dog.his.DAT
 ‘One can’t find any trace of Orbán’s dog.’
- (90) *Mindenhol azt olvasni, hogy a terhességre nem veszélyes a COVID-19 fertőzés.*
 everywhere that.ACC BE_ABLE_TO.find_by_reading-INF that the
 pregnancy.onto not dangerous the Covid-19 infection
 ‘One can find it written all over the place that Covid-19 is not harmful for pregnancies.’

⁵⁷ The classic example from the folk song: *Megismerni a kanászt ékes járásáról.* ‘One can recognize the swineherd by his proud gait.’

⁵⁸ The classic example from the folk song: *Oda megyünk lakni, ahol tejet kapni.* ‘We are moving there where one can find milk.’

2.3. Verbs of physical disposition

- (91) *Padovából könnyen elér-ni*
Padova.from easily PRT.BE_ABLE_TO.reach-INF
mind a tengert, mind a hegyeket.
both the sea.ACC both the mountains.ACC
'From Padova, one can easily reach both the sea and the mountains.'
- (92) *Nehezen bír-ni ezt a nagy meleget.*
with.difficulty BE_ABLE_TO.endure-INF this the big warmth.ACC
'It is difficult to cope with this heat.'
- (93) *[Ősszel Delhiben a hőség reggeltől délutánig tart, délután a lenyugvó nappal együtt a hőség is visszavonul.]*
este már el-bír-ni a zakót.
evening already PRT-BE_ABLE_TO.endure-INF the jacket.ACC
'[In Delhi, in the autumn, the heat lasts from the morning till the afternoon, in the afternoon, the heat recedes as the sun sets, and] in the evening, one can wear a jacket.'
- (94) *azért aszkéta én sem vagyok, ha a csajokkal este rágicsálunk egy kis mogyit, nem szenvedek... de egyedül nagyon könnyen ki-bír-ni!!!*
well ascetic I neither am if the girls.with evening chew.1PL a little nut.ACC not suffer.1SG but alone very easily PRT-BE_ABLE_TO.endure-INF
'I am not an ascetic myself, if we are having some snacks together with my friends, I do not torture myself, but on your own, you can withstand the temptation very easily.'
- (95) *Templomok, plébániák újulnak fel, szinte alig győz-ni a sok munkát.*
churches parishes new.MID.3PL PRT almost barely BE_ABLE_TO.cope_with-INF the much work.ACC
'Churches and parishes are getting renovated, one can hardly cope with all the related work.'
- (96) *Kis helyekre is könnyen be-fér-ni vele.*
small places.onto too easily PRT-BE_ABLE_TO.fit-INF it.with
'One can fit into small places too with it.'

3. Dispositional middles

Every sentence presented in this section is an actual attested sentence, either in the Hungarian National Corpus or in Hungarian-language electronic texts available on the internet.

3.1. Verbs of involuntary perception

- (97) *Hall-atsz-ik a tenger morajlása.*
BE_ABLE_TO.hear-MID-3SG the sea roar.3SG
'One can hear the roar of the sea.' = 'The roar of the sea is audible to everyone.'
- (98) *Innen lát-sz-ik a hegycsúcs.*
from.here BE_ABLE_TO.see-MID-3SG the summit
'One can see the summit from here.'
- (99) *Érz-őd-ik a virágok illata.*
BE_ABLE_TO.feel-MID-3SG the flowers scent.3SG

‘One can feel the scent of the flowers.’

3.2. Önkéntelen(fel)ismerés-igék

- (100) *Miről ismer-sz-ik meg egy jó cipő?*
 from.what BE_ABLE_TO.know-MID-3SG PRT a good shoe
 ‘How can one recognize a good pair of shoes?’
- (101) *Talán mindebből könnyebben ért-őd-ik meg,*
 perhaps from.all.this.more.easily BE_ABLE_TO.understand-MID-3SG PRT
ami most fog következ-ni.
 that.which now FUT.3SG follow-INF
 ‘Perhaps from this, one can understand more easily what is going to happen now.’
- (102) *A biztonságot sugárzó és nyújtó férfi*
 the security radiating and providing man
fel-ismer-sz-ik.
 PRT-BE_ABLE_TO.know-MID-3SG
 ‘One can easily recognize the kind of man who radiates and provides a sense of security.’
- (103) *Az idomok alatt valószínűleg környezetbarát (Euro-3)*
 the profiles under truly environmentally.friendly (Euro-3)
egyhengeres motor talál-sz-ik.
 single-cylinder engine BE_ABLE_TO.find-MID-3SG
 ‘Under the profiles one can find a truly environmentally friendly (Euro 3) single-cylinder engine.’
- (104) *Nehezen kap-ód-ik ember, mert*
 with.difficulty BE_ABLE_TO.find-MID-3SG man because
mind elementek Itáléba.⁵⁹
 all leave.PST.3PL Italy.to
 ‘One can barely find anyone, since everyone has left for Italy.’
- (105) *Szellemi tevékenységünk központja az agy szöveteiben*
 mental activities.1PL centre.3SG the brain tissues.3SG.INE
lel-edz-ik.
 BE_ABLE_TO.find-MID-3SG
 ‘The centre of our mental activities can be found in the tissues of the brain.’
- (106) *%Nem minden csacskaóhyan csacska, amilyenek*
 not every silliness so silly such.DAT
olvassz-ik.
 BE_ABLE_TO.find_by_reading-3SG
 Not every piece of stupidity is as stupid as it can appear upon reading.

4. As ability modal auxiliaries

Every sentence presented in this section is an actual attested sentence, either in the Hungarian National Corpus or in Hungarian-language electronic texts available on the internet.

- (107) a. *Félig-meddig meg-vakul-t-am, nem lát-ok*
 sort.of PRT-blind_v-PST-1SG not be_able_to_{visual}-1SG
olvas-ni, ír-ni.

⁵⁹ Moldavian Hungarian (https://www.telekialapitvany.hu/wp-content/uploads/2013/02/moldvai_csangok_es_a_valtozo_vilag.pdf)

- read-INF write-INF
 ‘I am half blind, I can’t read or write (due to a degradation of my ability to see).’
- b. *Ha teljesen sötét van a szobában, akkor nem lát-ok ír-ni a laptop billentyűzetén.*
 if completely dark is the room.in then not
 be_able_to_{visual}-1SG write-INF the laptop keyboard.3SG.on
 When it’s completely dark in the room, I cannot type on the laptop (due to my inability to see clearly).
- (108) *Másfél óra után már nem bír-ok figyel-ni az előadásra.*
 1.5 hour after already not be_able_to_{endurance}-1SG
 pay.attention.to-INF the talk.onto
 After one and a half hours, I can no longer pay attention to the talk (due to my lack of mental endurance/stamina).
- (109) *A szállításnál alaposan becsomagolták, minden sértetlenül érkezett, alig győző-t-em kicsomagol-ni.*
 the transportation.at carefully pack.PST.3PL
 everything intact arrive.PST.3SG
 barely be_able_to_{capacity}-PST-1SG unpack-INF
 ‘The delivery company packed it very carefully, everything arrived intact, I was almost unable to unpack it.’
- (110) a. *Ha esetleg az érdeklődésedet véletlenül fel talál-t-am kelt-eni*
 if maybe the interest.your.ACC accidentally
 PRT be_able_to_{accidental}-PST-1SG raise-INF
 ‘If I accidentally managed to arouse your interest.’
- b. *Ne vitatkozz velem, mert meg talál-lak harap-ni*
 not argue.IMP.2SG with.me because
 PRT be_able_to_{accidental}-1SG bite-INF
 ‘Don’t you keep arguing with me, or I might bite you (impulsively).’
- (111) *Földrajzból nem ért-em meg-csinál-ni*
 geography.from not be_able_to_{cognitive}-1SG PRT-do-INF
 a házi, kérhetek segítséget?
 the homework.ACC ask.MAY.1SG help.ACC
 ‘I am unable to do the geography homework (due to my inability to understand the task), can I ask for a bit of help?’
- (112) a. *Olvas-t-am az előző üzidet is, csak tegnap nem ér-kez-t-em válaszol-ni.*
 read-PST-1SG the previous message.your.ACC too
 only yesterday not BE_ABLE_TO.reach-MID-PST-1SG answer_v-INF
 ‘I have read your previous message too, I just could not answer yesterday (due to an inability to fit it into my packed schedule).’
- b. *Még nem ér-t-em el-olvas-ni, kb. addig jut-ott-am, hogy át tud-t-am lapoz-ni.*
 yet not be_able_to_{scheduling}-PST-1SG PRT-read-INF appr. that.til
 get-PST-1SG that PRT be_able_to-PST-1SG leaf_v-INF
 ‘I did not manage to read it yet (due to my packed schedule), I could just leaf through it for the time being.’
- c. *János meg-ér-t-e be-hoz-ni*
 John PRT- be_able_to_{scheduling}-PST-3SG PRT-gather-INF

*az összes tűzifát.*⁶⁰
 the all firewood.ACC

⁶⁰ Western Transdanubia (p.c. András Cser).